The Hostel Laundry Queue

Week 8 Coding Assignment - Question 1

Design: Student 1--0..* WashCycle; WashingMachine 1--0..* historical WashCycle and at most one active cycle.
WashCycle references one Student, one WashingMachine, and one WashType. WashType provides duration and charge.
Java solution:

```
interface WashType { String name(); int durationMinutes(); double charge(); }
class Quick implements WashType { public String name(){return "Quick";} public int durationMinutes(){return 30;} public double charge(){return 20;} }
class Normal implements WashType { public String name(){return "Normal";} public int durationMinutes(){return 45;} public double charge(){return 30;} }
class Heavy implements WashType { public String name(){return "Heavy";} public int durationMinutes(){return 60;} public double charge(){return 45;} }
class Student { final String name; Student(String n){name=n;} }
class WashCycle { final Student student; final WashType type; WashCycle(Student s,WashType t){student=s;type=t;} }
class WashingMachine {
private final String id; private WashCycle active;
WashingMachine(String id){this.id=id;} boolean isFree(){return active==null;}
String start(Student s,WashType t){
if(active!=null)return "Machine "+id+" is currently busy.";
active=new WashCycle(s,t);
return t.name()+" wash started on "+id+" for "+s.name+" ("+t.durationMinutes()+" min). Charge: Rs "+String.format("%.2f",t.charge())+".";
}
String complete(){if(active==null)return id+" has no active cycle.";active=null;return id+" cycle completed. "+id+" is now free.";}
}
public class Main { public static void main(String[] args){
WashingMachine m1=new WashingMachine("M1"),m2=new WashingMachine("M2");
Student asha=new Student("Asha"),ravi=new Student("Ravi"),neha=new Student("Neha");
System.out.println(m1.start(asha,new Quick()));
System.out.println(m1.start(ravi,new Heavy()));
System.out.println(m2.start(ravi,new Heavy()));
System.out.println(m1.complete());
System.out.println(m1.start(neha,new Normal()));
} }
```

Expected output (currency shown as Rs to keep source code ASCII):

```
Quick wash started on M1 for Asha (30 min). Charge: Rs 20.00.
Machine M1 is currently busy.
Heavy wash started on M2 for Ravi (60 min). Charge: Rs 45.00.
M1 cycle completed. M1 is now free.
Normal wash started on M1 for Neha (45 min). Charge: Rs 30.00.
```

Extensibility: add Delicate by implementing WashType. Keep active-cycle state private.